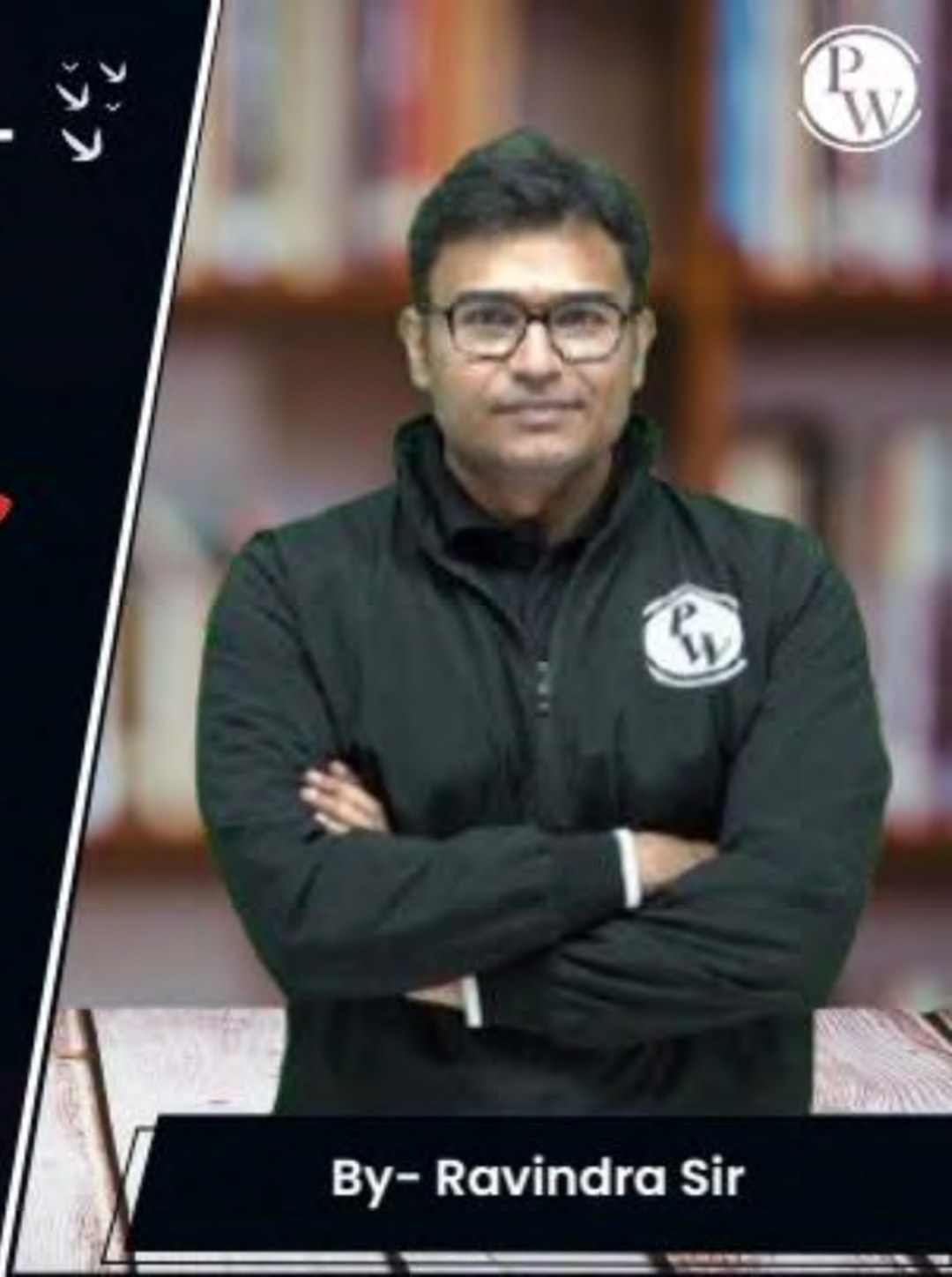
Computer Science & IT

Theory of Computation



Turing Machines

Lecture No. 4



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

TM

Topics to be Covered



- Topic TM
- Topic
- Topic

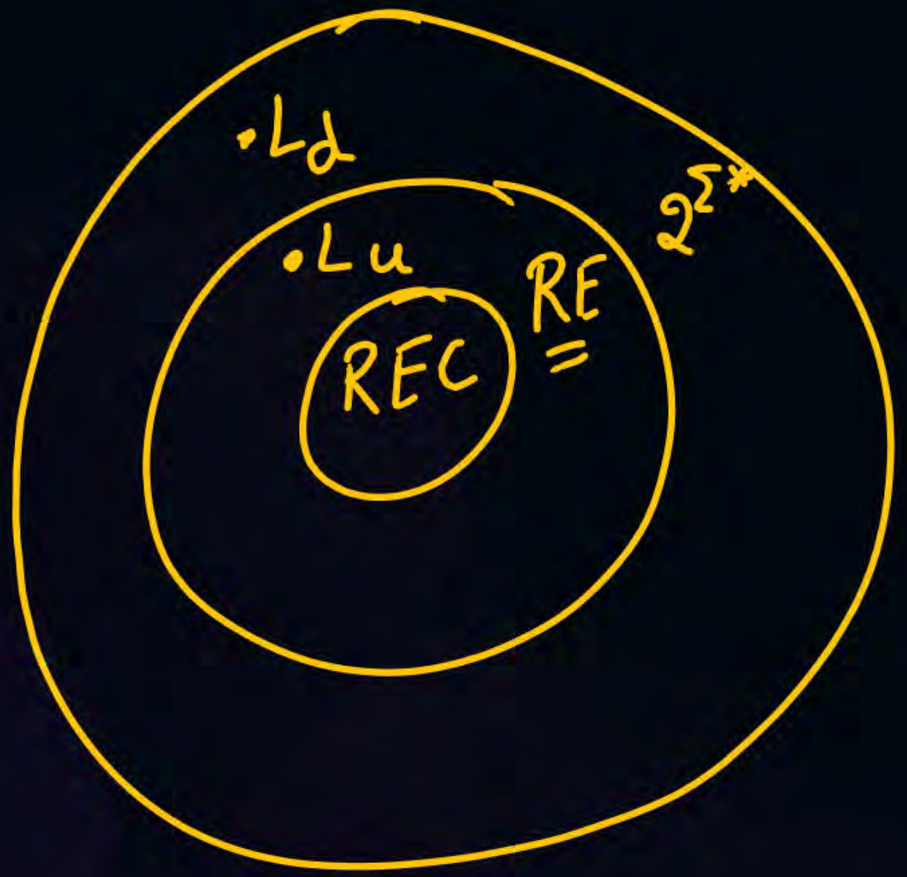
$L_u \rightarrow$ universal language of TM

$$L_u = \{ e(T) / e(T) \in L(T) \}$$

$e(T)$ ↓ encoding in (0, 1)

$L_d \rightarrow$ diagonalisation of TM

$$L_d = \{ \underline{\underline{e(T)}} \mid \underline{\underline{e(T)}} \notin \underline{\underline{L(T)}} \}$$


L_d
L_u
REC
RE
=
2^{Σ^*}

1) Halting problem -

Given a TM and $w \in \Sigma^*$ will TM halt on w. This is undecidable.

Blank Tape halting problem :-

Given a BT, to a TM, will it halt

It is undecidable

State entry problem:-

Given a TM, will it enter a particular State

undecidable

Post Correspondence problem (PCP) :-

Given $(u_1, u_2, u_3 \cdots u_n)$ and $(v_1, v_2, v_3 \cdots v_n)$

Some combination of u should be equal to some combination of v.

$u_3 u_1 u_4 = v_3 v_1 v_4$ This undecidable

modified PCP:-

It is same as PCP. But starting should be u_1, v_1

$u_1 u_3 u_5 = v_1 v_3 v_5$

$u_1 u_5 u_4 = v_1 v_5 v_4$

it is undecidable

RE membership :-

undecidle

Rice's theorem :-

Every non trivial question on (RE) language is undecidable

no trivial ques ⟵ (Yes or NO)

Trivial :- ⟶ always yes or always no

Reduction theorem :-

$L_1 \leq L_2$ (L_1 is reducible to L_2)

It means that every string in L_1 can be converted to string in L_2 in polynomial time

if $L_1 \leq L_2$

UD → UD

D ← D

D → no comment

UD ← no comment

REC ← REC

Not REC ← Not REC

$L_1 \leq L_2$

SD ← SD

RF ← RF ✓

Not SD ← Not SD ✓

NOT RF → Not RF

✓

If $L_1 \leq L_2$ and L_1 is UD; then what is $L_2 \rightarrow$ UD

if $L_1 \leq L_2$ and $L_2 \leq L_3$

$L_2 = RE$, then $L_1 = ?$ $L_1 \rightarrow RE$

$L_3 = ?$ $L_3 \rightarrow$ no comments

$L_1 \leq L_2$ and $L_2 \leq L_3$

UD → UD UD → UD

$L_1 \to UD$

$L_2 = ?$

$L_3 = ?$

$L_1 \leq L_2$

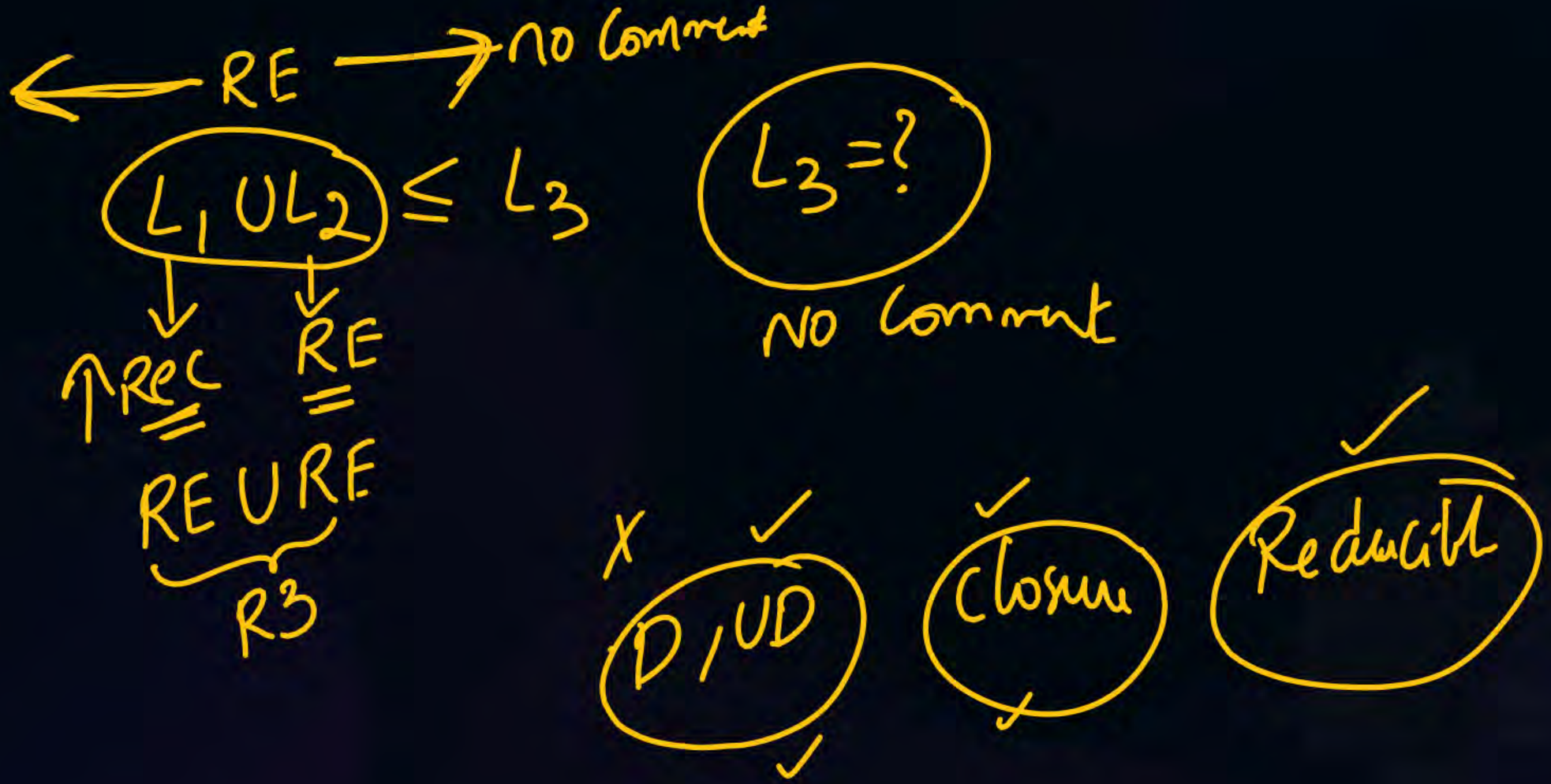
if L_2 in Rec, then L_1 in RE. (T/F)

(T)

$L_1 \leq L_2$

Rec ← Rec

Every Rec in (RE)


RE
No Comment
$L_1 \cup L_2 \leq L_3$
$L_3 = ?$
No Comment
Rec
RE
RE ∪ RE
R3
X
✓
D / UD
✓
Closure
✓
Reducible

→ L is Rec iff both L and $\bar{L}$ has turing enumerable procedure

→ Some language is turing enumerable but its complement has no turing enumeration then language is RE but no Rec.



→ L is rec iff it has membership algo.

$w \overset{?}{\in} L$ < Y / N

→ L is Rec if L can be enumerated in lexicographic ordering.

↓

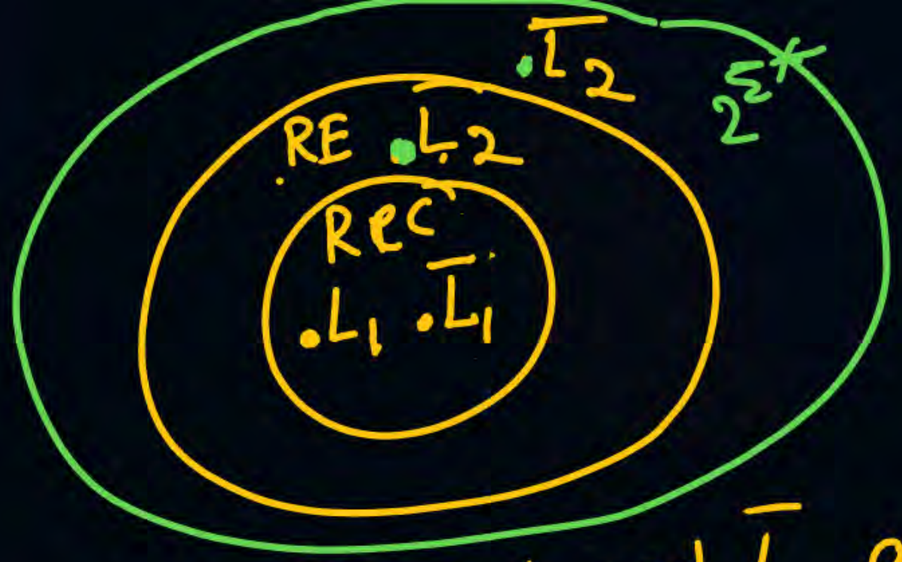
dictionary ε, a, b, aa, ab, ba, ...

Closure property :-

$\cup, \cap, \cdot, *$ → RE and REC closed

L is Rec iff $\bar{L}$ is Rec

L is RE ⇒ $\bar{L}$ may / may not be RE



$L, \bar{L}$ theorem :-

① if L and $\bar{L}$ are both RE $\Rightarrow$ L and $\bar{L}$ are REC

② if L is RE but not Rec, then surely $\bar{L}$ is not RE

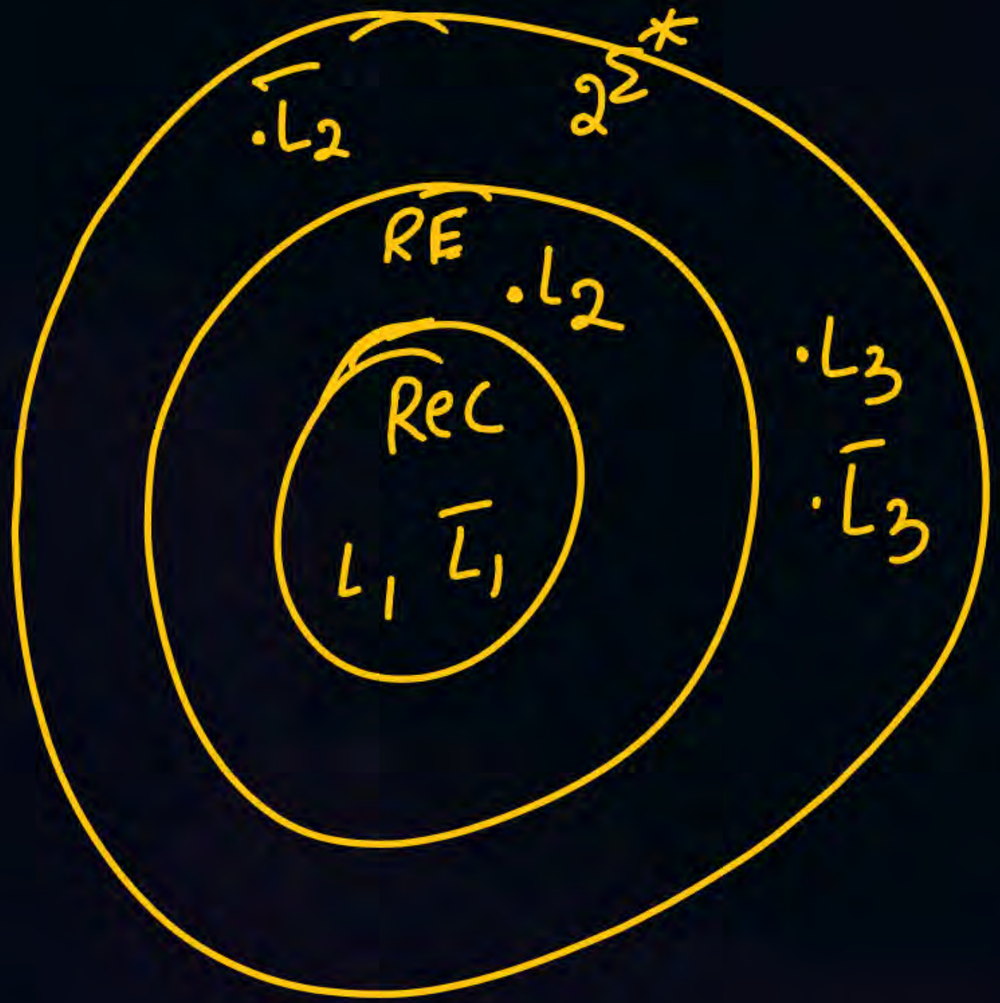


③ If L and $\bar{L}$ are Complementary languages

✓ I) Both are Rec

II) One is RE but not Rec, another is not RE

III) Both are not RE


$\overline{L_2}$
2^{Σ^*}
RE
L_2
Rec
L_1 $\overline{L_1}$
L_3
$\overline{L_3}$

L is REC $\rightarrow$ $\bar{L}$ is Rec

L is RE but not Rec $\rightarrow$ L is not RE

L is not RE $\Rightarrow$ $\bar{L}$ may or may not be RE

L is not REC $\Rightarrow$ $\bar{L}$ is not REC

Countability :-

- Set
  - ∞
    - uncountably ∞
    - countably ∞
  - Finite

A Set 'S' is Countable iff S has a enumeration procedure.

EP means a procedure which lists a set with a guarantee that any particular element will be listed in a finite amount of time

+1

1, 2, 3, 4, 5, ... (0.1)

0, 0.1

Natural numbers :- (Countably ∞)

+1

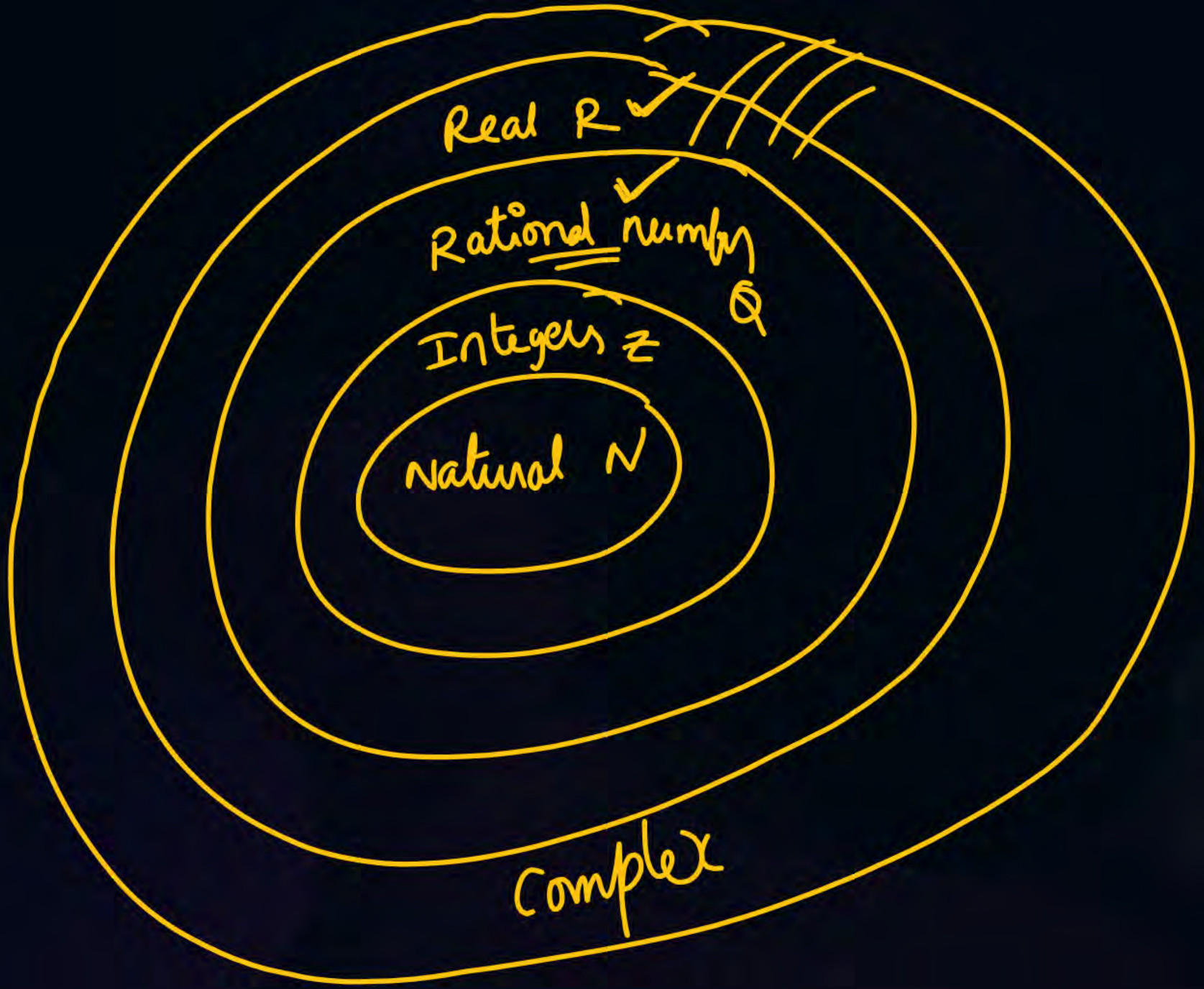
1, 2, 3, 4, 5, 6 ..... $10^{10^{10}}$

Real numbers :- +0.1 (+0.01) (uncountably ∞)

0, 0.1 0, 0.01

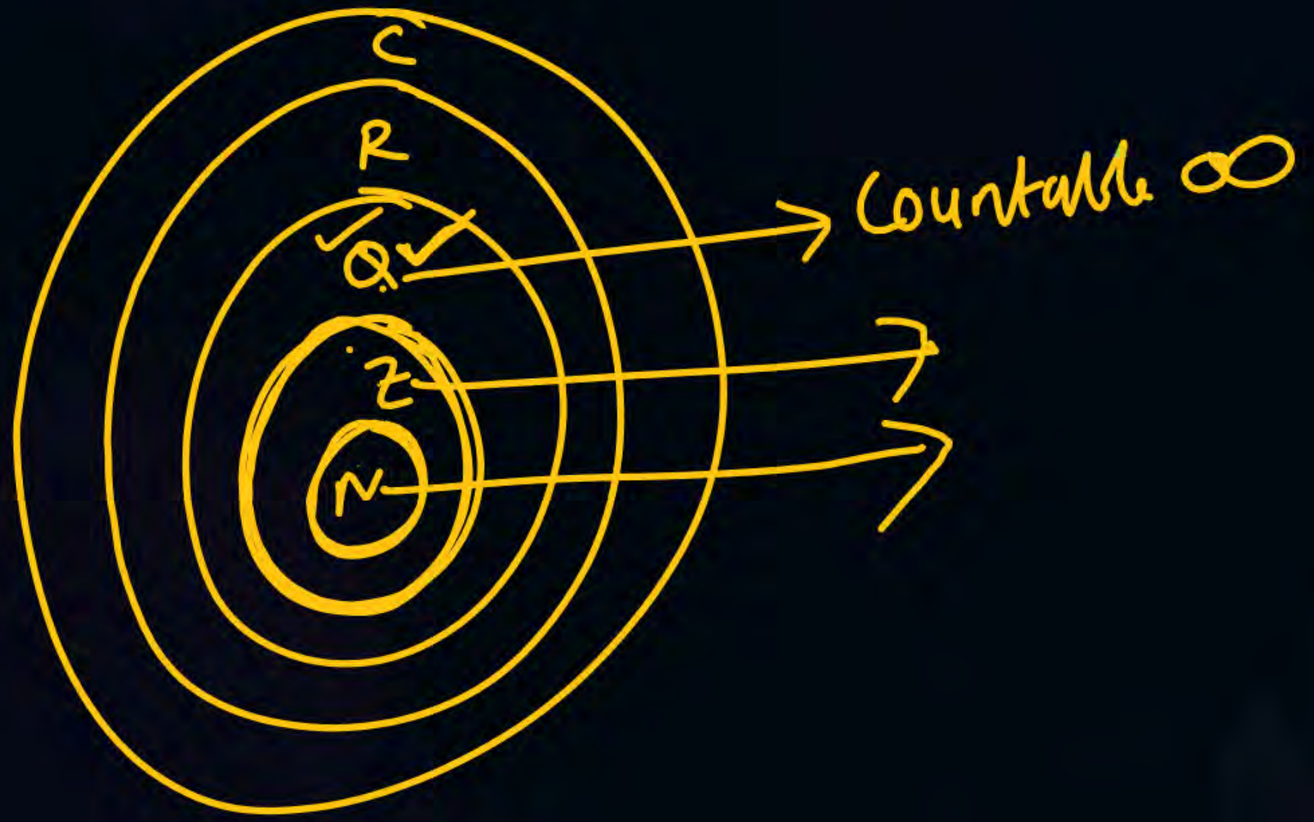
0.01 ✗ 0.001✗

0.001✗ 0.001✗


Real R
Rational number
Q
Integers Z
Natural N
Complex

Natural, Integers, Rational → Countably ∞

Real, Complex → Uncountably ∞


C
R
Q
Z
N
Countable ∞

1/1+1 1/2+1

Rational P/q :- → enumeration

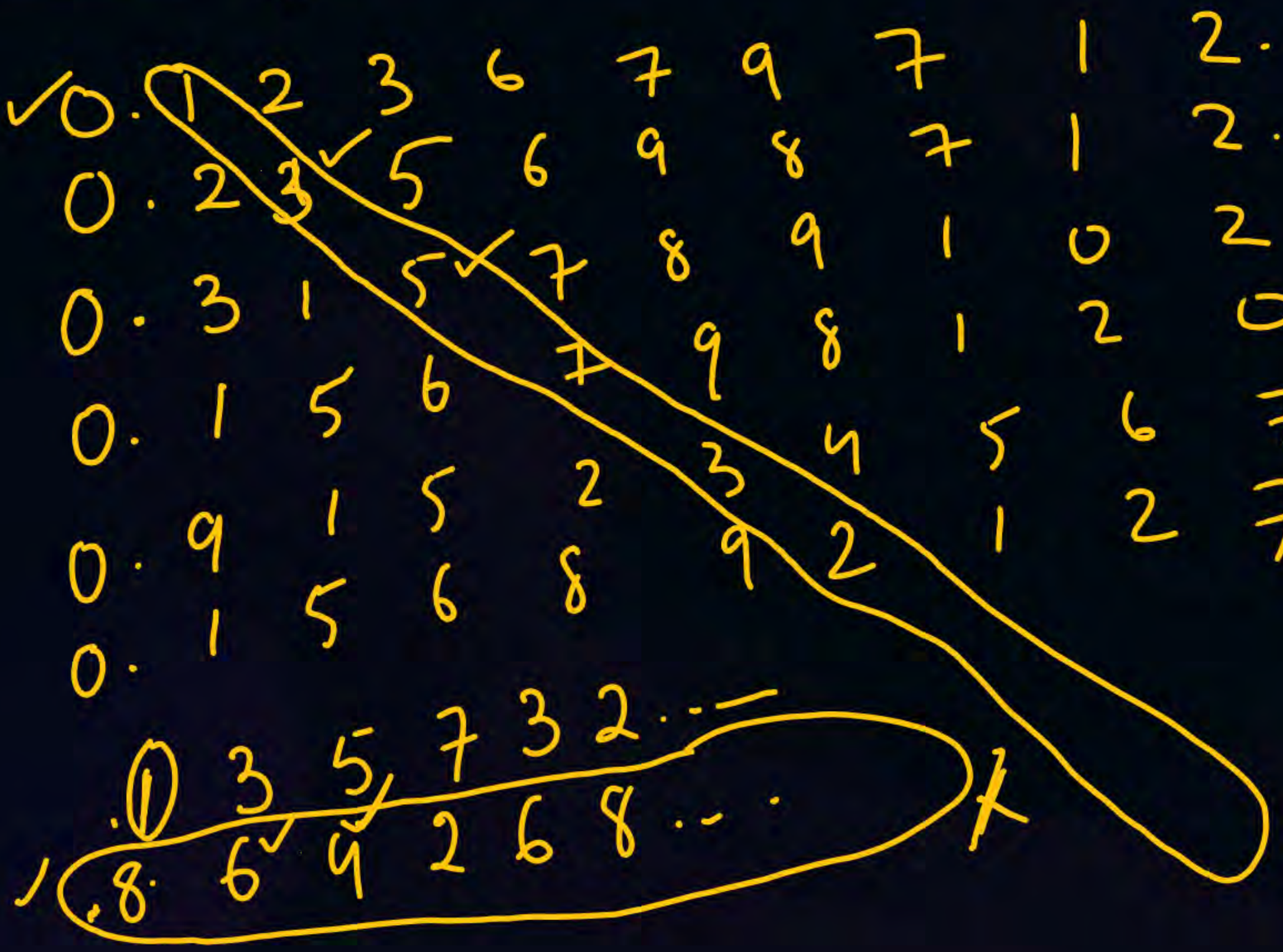
1/1	1/2	1/3	1/4	1/5	.....
2/1	2/2	2/3	2/4	2/5	----
3/1	3/2	3/3	3/4	3/5	----
4/1	4/2	4/3	4/4	4/5	----

if $A \subseteq B$, B is countably ∞, A is countably ∞ or finite.

Q is countably infinite → Z, N, whole, prime, composite,
even, odd → Countable ∞

↓
Rational numbers

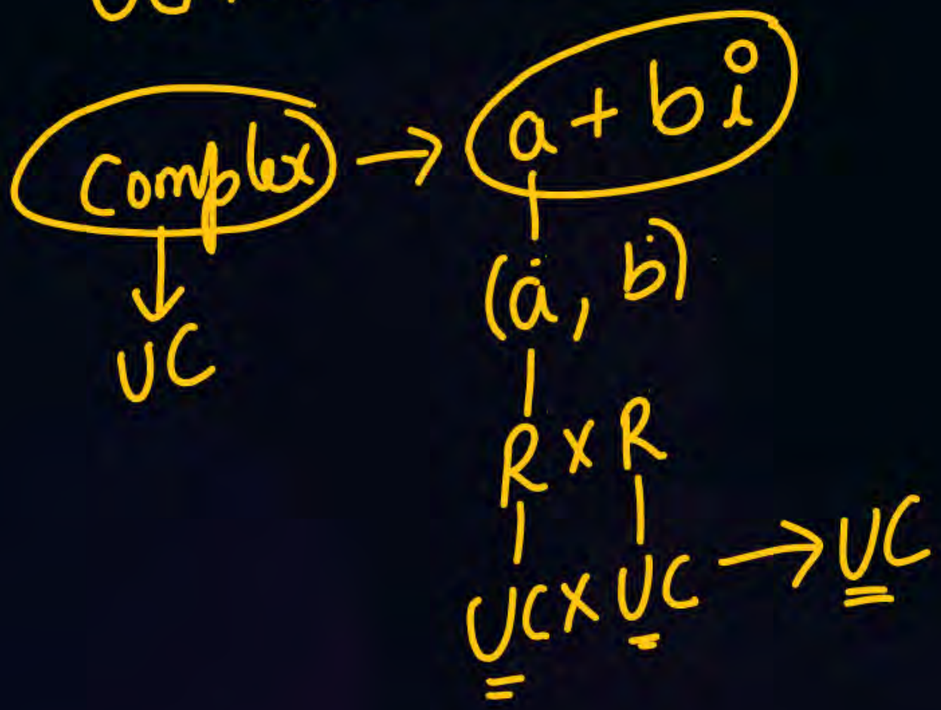
Cantor's diagonalisation:- We can prove real number an uncountably infinite

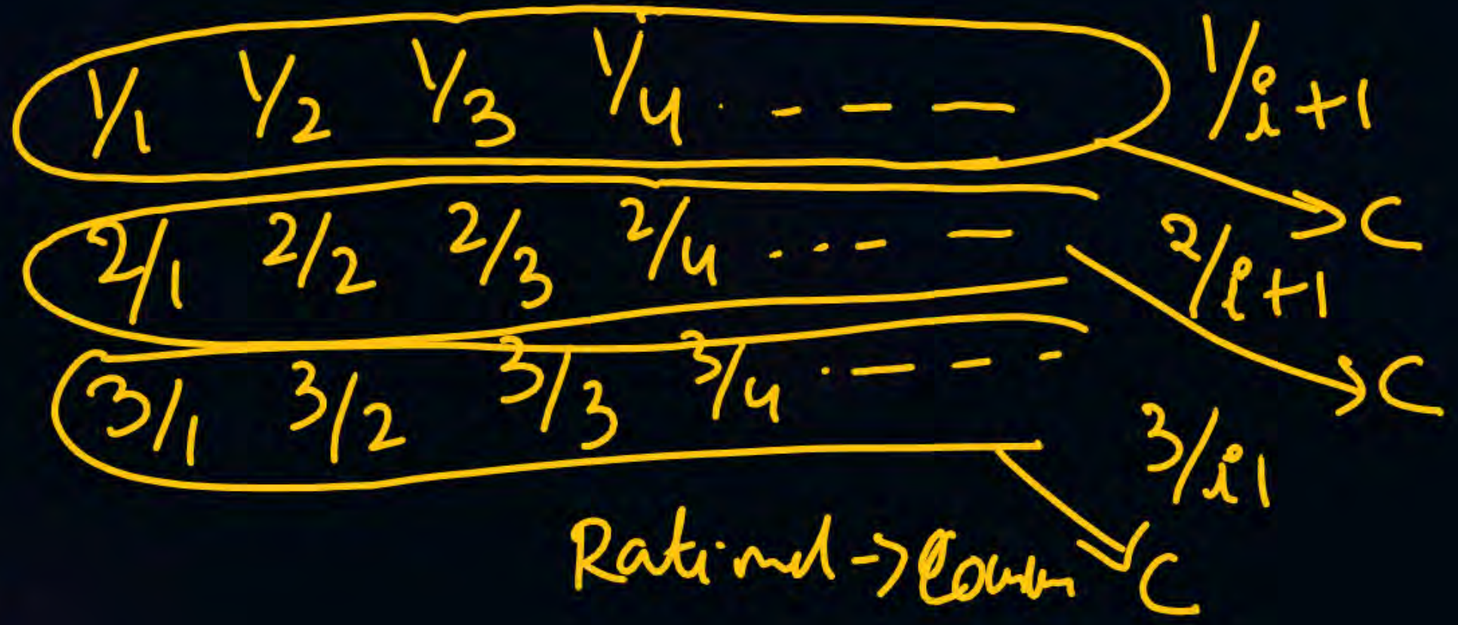


Real → uncountably ∞

$C \times C = C$

$UC \times UC = UC$





$UC \times C = UC$

$C \cup C = C$ ✓

$C \cup C \cup C \cup C \cdots\cdots = C$ ✓

$(UC) \cup (UC) \rightarrow UC$

$(UC) \cup (C) \rightarrow UC$

Cantor's theorem :-

if S is CI $\rightarrow 2^S$ is UCI

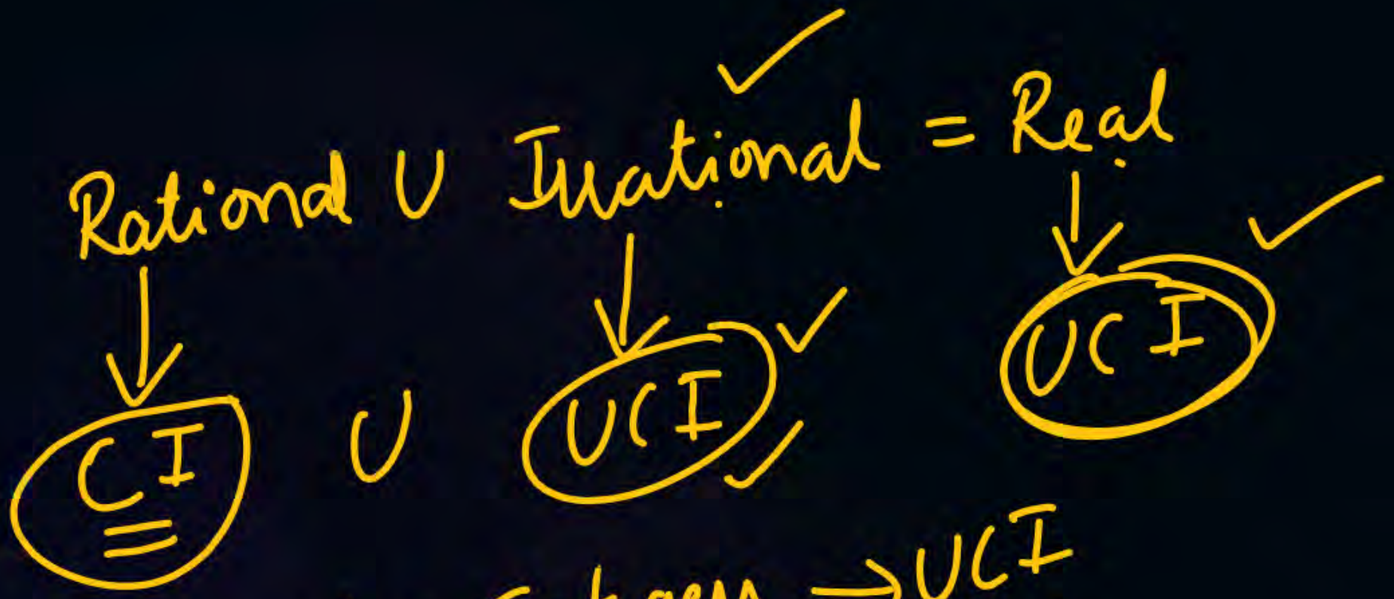
if S is Finite $\rightarrow 2^S$ is finite

N is CI $\Rightarrow 2^N$ is UCI

Z is CI $\Rightarrow 2^Z$ is UCI

Q is CI $\Rightarrow 2^Q$ is UCI

Rational ∪ Irrational = Real



Non Integers → UCI

Non Rational → UCI

TOC Sets :-

Σ^* → enumeration procedure → proper ordering

↓

Countably ∞

$\Sigma = \{a, b\}$

length wise, lexicographic

$\{0, a, b, aa, ab, ba, bb, ____$

$\emptyset$

Σ^* is CI

$L_1 = \text{even len}$ $\quad L \subseteq \Sigma^*$ ✓

CI $\quad$ CI $\quad$ CI

$L_2 = a^n b^n$

CI

$L_3 = a^n b^n c^n$

CI

2 min

Every language is countably infinite

Set of all Neg languages - CI

Set of all CFLs - CI

Set of all CSLs -> CI

Set of all Rec - CI

Set of all RE - CI

TM → encode → (0, 1)

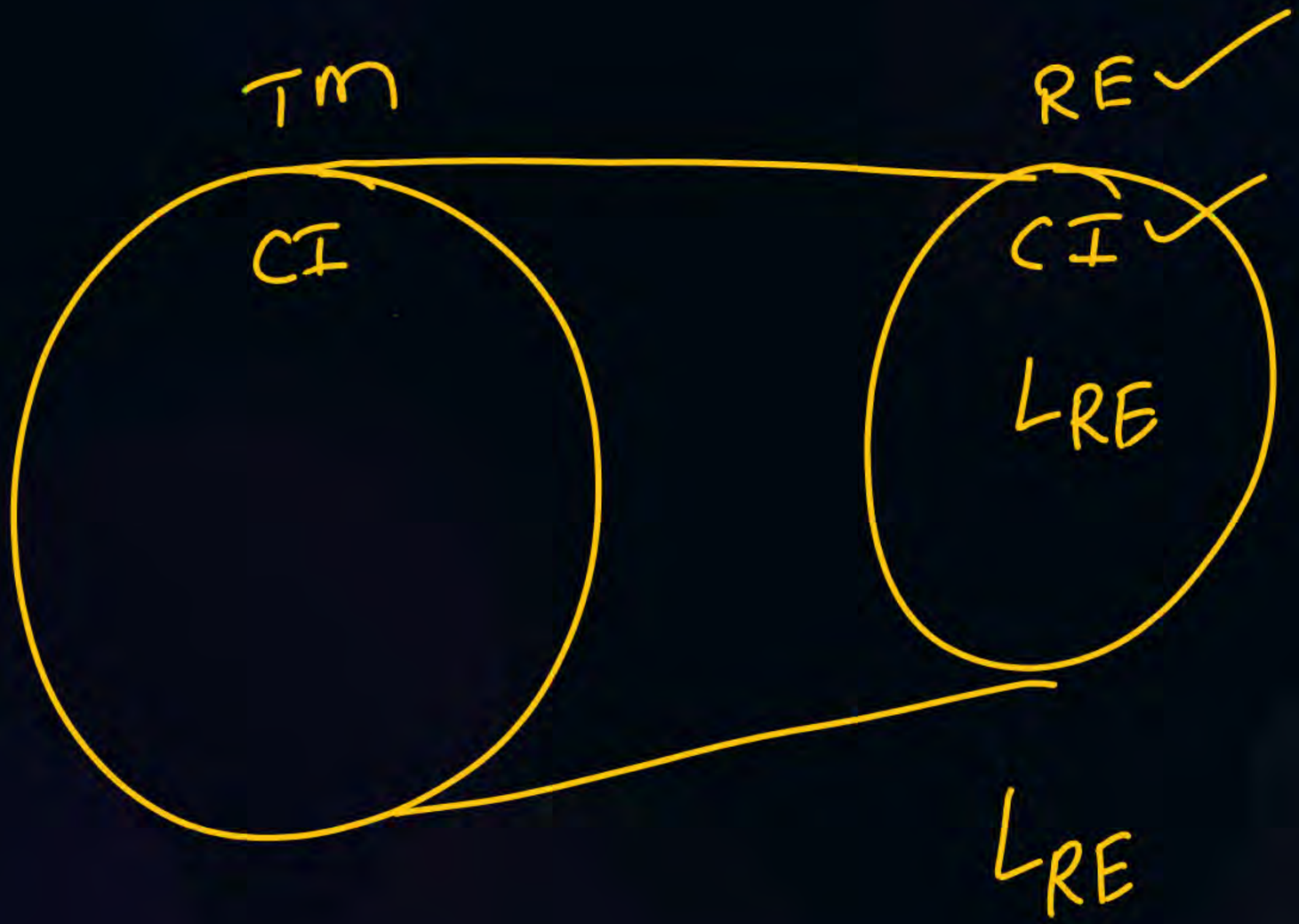
TM → String of (0, 1)

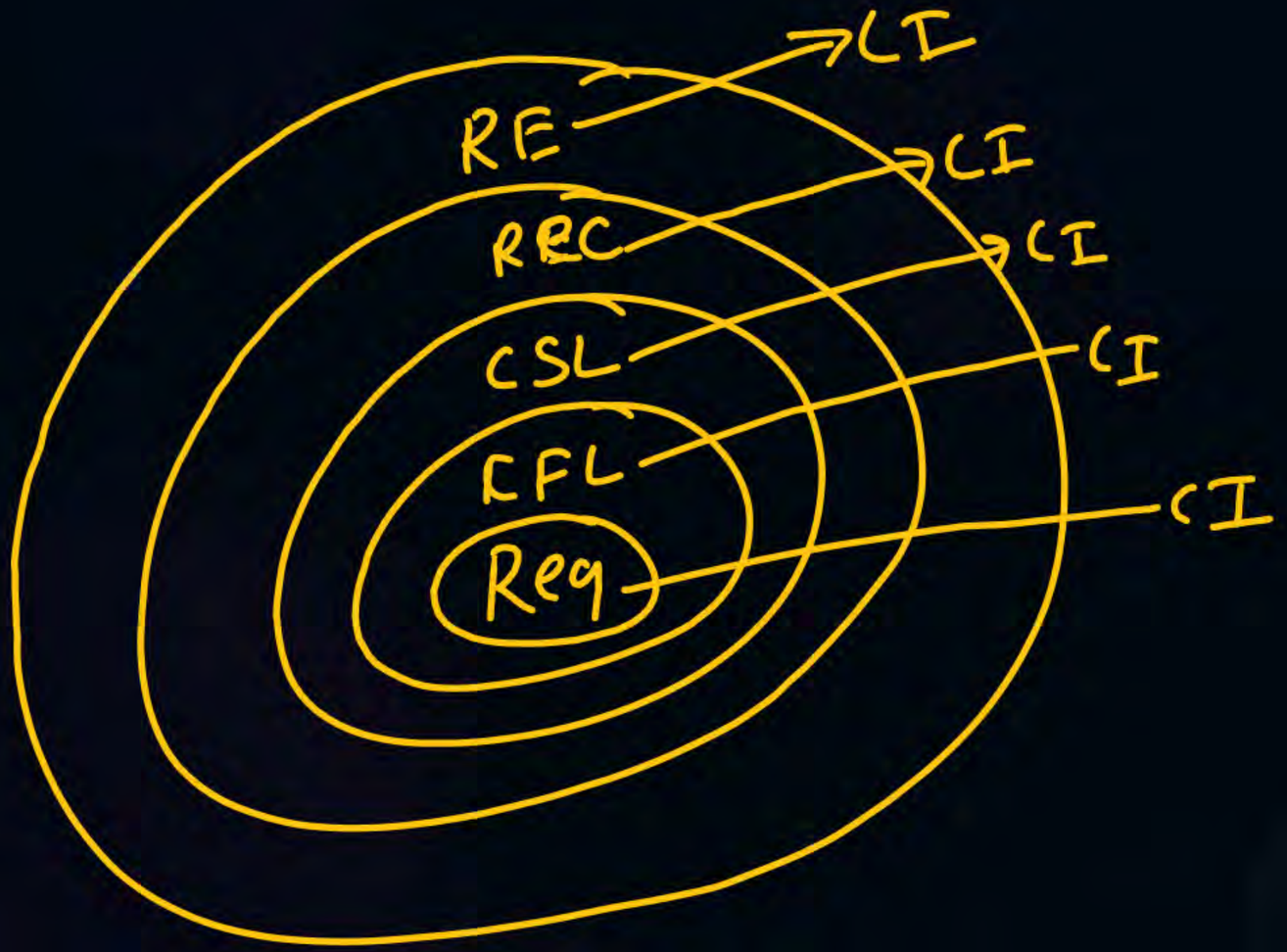
Set of all TM's is a language

Set of all TM's ⊆ Σ^* ✓

Set of all TM's → CI

Σ^* → CI ✓


TM
RE
CI
CI
LRE
LRE


RE
RRC
CSL
CFL
Reg
CI
CI
CI
CI
CI

Σ^* is CI → 2^{Σ^*} is UCI ✓

Set of all languages



L_{RE} → CI

$U L_{not\,RE}$ → 2^{Σ^*} → UCI

$U L_{not\,RE}$ → 2 → UCI

	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	✗	✓	✓	✓	✓
$\cap$	✓	✗	✗	✓	✓	✓
L^*	✓	✓	✗	✓	✓	✗
$*$	✓	✗	✓	✓	✓	✓
$\cdot$	✓	✗	✓	✓	✓	✓
L^R	✓	✗	✓			
$h(L)$	✓	✗	✓			
$h^{-1}(L)$	✓	✓	✓			


AIR
River
ALL INDIA RANK


L
Reg
DCFL
CFL
CSL
REC
RE
L ∪ R
L ∩ R
L - R
↑L



2 mins Summary

Topic

Turing Machines

Topic

Topic

Topic

Topic

THANK - YOU